

Sarthak Kakkar

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Computer Science Student at Northeastern | AI/LLMs • Full-Stack Development • Embedded Systems

EDUCATION

Bachelor of Science in Computer Science | Business Administration minor

Northeastern University, Khoury College of Computer Science | Boston, MA | GPA: 3.72 | 08/2026

WORK EXPERIENCE

Fresenius Medical Care | Lawrence, USA | 01/2026 - 06/2026

Computer Science / Software Engineer Co-op | Full-time

- Built a high-accuracy **RAG knowledge system** on **Azure** over clinical and technical healthcare documents using specialized retrieval pipelines.
- Architected a **modular pipeline** to compare multiple RAG configurations, improving retrieval performance through evaluation metrics.
- Introduced agentic workflows like **LangGraph** for controlled, context-aware interactions and patient-facing AI systems.
- Collaborated with stakeholders, domain experts, and users to curate evaluation datasets and support **production deployment**.

Movement Neuroscience Lab | Boston, USA | 01/2025 - 06/2025

Embedded Systems Engineer Co-op | Full-time

- Engineered **C++/Python** embedded system with electrical stimulators & force sensors, enabling **IRB-approved** human subject experiments.
- Re-architected a vendor hardware API into an **asynchronous, modular system**, enabling real-time stimulation control across **3+ projects**.
- Reverse-engineered undocumented BLE hardware and created **Python APIs** for real-time data streaming and device control.
- Automated end-to-end pipelines from raw sensor input to **clinician-ready visualizations**, producing **research-grade** reports for federal trials.

Northeastern University | Boston, USA | 09/2024 - 12/2024

Teaching Assistant - Programming in C++ | Part-time

- Taught over 100 students advanced **C++** (OOP, memory management, smart pointers), optimizing course-wide **Linux** development workflows.
- Delivered independent lecture on **smart pointers** with live coding demos, simplifying complex **memory-safe design** for large student audiences.
- Debugged and optimized **student projects**, resolved Khoury **Linux server** issues, and streamlined productivity across multiple course sections.

Wissen Infotech | Bangalore, India | 06/2024 - 08/2024

LLMOps Intern | Full-time

- Built **Python code-conversion accelerator** projected to reduce enterprise migration timelines by **2–3 months** across global client systems.
- Designed **LangChain/LangGraph** feedback loop for autonomous error detection, targeted regeneration, and recovery, boosting pipeline efficiency.
- Drove **LLM adoption** by delivering technical prototypes/demos to executives and prospective clients, influencing high-level business strategy.
- Benchmarked **OpenAI, DeepSeek, and GenAI models**; authored comprehensive reports that guided internal adoption and client recommendations.

Applify Tech Private Limited | Mohali, India | 07/2023 - 08/2023

Intern | Full-time

- Developed **Java backend services** with **REST APIs**, **SQL** queries, and **JDBC** integration, strengthening scalability and production reliability.
- Implemented **JaCoCo unit tests** and **debugged** production code, improving release stability and reducing runtime issues.
- Collaborated in an **agile** environment using **Git** and **JIRA**, documenting workflows and aligning sprint goals with client needs.

PROJECTS

PawtoGrader Platform

- Served as **Technical Lead and full-stack developer**, driving development with **TypeScript/Supabase/Docker** and owning project deliverables.
- Selected for an exclusive practicum project, **leading a 6-person Agile team** to extend PawtoGrader for **3k+ Khoury College CS students**.
- Delivered new features including **surveys, dashboards, and automated group formation**, improving platform usability and scalability.

Retrieval as a Service

- Built and open-sourced **RaaS**, a self-hosted retrieval layer for RAG workflows with **project-scoped APIs** and **Chroma** indexing.
- Developed a **FastAPI + Next.js** system supporting document upload, ingestion, and secure per-project API key access.
- Designed a modular retrieval backbone with **Dockerized deployment** enabling reusable, scalable RAG pipelines without repeated setup.

AI Powered Communication Assistant

- Developed an **AI-powered communication assistant** using LangChain, LangGraph, and LangServe to automate professional inquiry handling.
- Deployed a **Dockerized FastAPI backend** on **AWS App Runner** for scalable, stateless processing.
- Integrated a **React + Vite frontend** hosted on **Netlify**, enabling secure, real-time interaction and notification delivery.

Squegg Python API

- Reverse-engineered the **BLE protocol** of the Squegg smart squeeze ball, enabling seamless **desktop integration** for analytics and control.
- Built open-source **Python API** streaming grip strength, battery status, and notifications, delivering reliable real-time device monitoring.
- Packaged tool with **PyInstaller** into a cross-platform executable, deployed in **medical research labs** for clinician-ready data access.

Typographic Latent Space (ICLR 2023 Publication)

- Co-authored **ICLR 2023 paper** mapping digit latent space using **β-VAE**, advancing generative design and representation research.
- Implemented automated **PyTorch training loops** with reproducible configs, metadata, and labeled sample grids for validation.
- Explored latent structure through **controllable sampling of generative factors**, revealing design patterns and model transparency.

Portfolio Website

- Built a **responsive** portfolio with **React**, **React Router**, and **Tailwind CSS**, featuring polished **GSAP animations** and smooth transitions.
- Implemented **dark/light mode** and modular **JSON-based content architecture**, ensuring fast updates and scalability.
- Deployed on **Netlify** with continuous integration and automated Git builds, streamlining deployment and publishing workflows.

EcoSim: Dynamic Predator-Prey Ecosystem Simulation

- Developed **C++ OOP simulation** (inheritance, polymorphism) to model predator-prey dynamics with accurate ecosystem variability.
- Added **critter behaviors** simulating realistic movement, interactions, and emergent environmental complexity.
- Deployed and fine-tuned simulation on **Khoury Linux servers**, ensuring performance stability and showcasing scalable **C++ design**.

Snake Game

- Developed a Snake Game with a Command Line Interface using the Ncurses library in C++.
- Introduced dynamic gameplay with randomly placed, size-varying food items, evolving obstacles, and adaptive difficulty.
- Deployed on the Khoury Linux Server with version control and collaboration through GitHub.

Calendar App

- Developed a Java-based calendar application with a weekly view, secured access via password, a progress tracker, and the classification of tasks and events.
- Configured the application with a graphical user interface using JavaFX that allows customizable themes.

Battleship Game

- Created a Battleship game in Java with player-vs-algorithm and automated algorithm-vs-algorithm modes.
- Structured the application with JSON data handling and the Proxy Design Pattern for maintainability.
- Configured Gradle build automation with JaCoCo integration, achieving over 90% unit test coverage.

PUBLICATIONS

Mapping the Typographic Latent Space of Digits | 04/2023

International Conference for Learning Representations

Used disentangled Beta-VAE's in an unsupervised learning approach to map latent feature spaces with a dataset of MNIST Style Typographic Images across 2990 unique font styles, helping typographers explore new attributes for their classification systems.

SKILLS

Programming Languages: C++, Java, JavaScript, OCaml, Python, SQL, TypeScript

Frameworks & Libraries: LangChain, LangGraph, Matplotlib, NumPy, Pandas, PyTorch, React, scikit-learn, Tailwind CSS

Systems & Tools: Git, GitHub, JIRA, Linux, Netlify, Streamlit

Software Development: Agile, API Development, Async Programming, CI/CD, Docker, E2E/Unit Testing, Object-Oriented Design (OOD), PostgreSQL

Machine Learning & Data Science: Agentic AI Development, Data Visualization, Deep Learning, Supervised Learning, Vectorization

Embedded Systems: Arduino, Asynchronous & Event-Driven Architectures, BLE, Raspberry Pi, USB HID, Wireless & IoT Integration

Frontend & Web Development: HTML5, Next.js, Playwright Testing, React, React Router DOM, Server-Side Development, Tailwind CSS

Backend Development: Authentication & Authorization, Database Integration, Express.js, FastAPI, RESTful API Design, Server-Side Development